(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JULY 2023

Paper Code: ES-108 ***Subject: Electrical Science***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

Q1 a) Defining the following terms: **[3]**
a. Active Element, b. Passive Element, c. Circuit vs. Network

b) What is resonance in the AC circuit? How is it generated in series and parallel circuit? **[3]**

c) What are the types of torques in Indicating Type measuring instruments? **[3]**

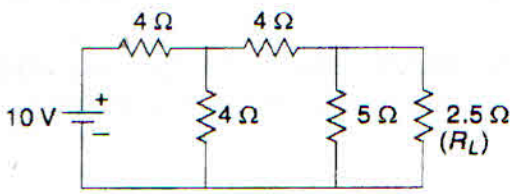
d) Interpret the different types of losses in the 1ϕ Transformer **[3]**

e) List out the application of DC Motor and generator? **[3]**

UNIT-I

Q2 a) State and proof the Maximum Power Transfer theorem having efficiency 50% with suitable expressions and diagram. **[7.5]**

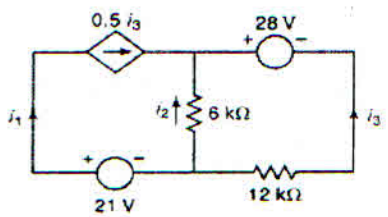
b) Evaluate the current through load resistance RL using Norton's Theorem. **[7.5]**



OR

Q3 a) Distinguish between dependent and independent sources. How do you transform a voltage source into a current source? **[7.5]**

b) Calculate *i1, i2,* and *i3* in the given circuit **[7.5]**



UNIT-II

Q4 a) Illustrate the basic terminology used in AC Systems with the help of waveforms? Also, Differentiate single phase and three phase circuit. **[7.5]**

P.T.O.